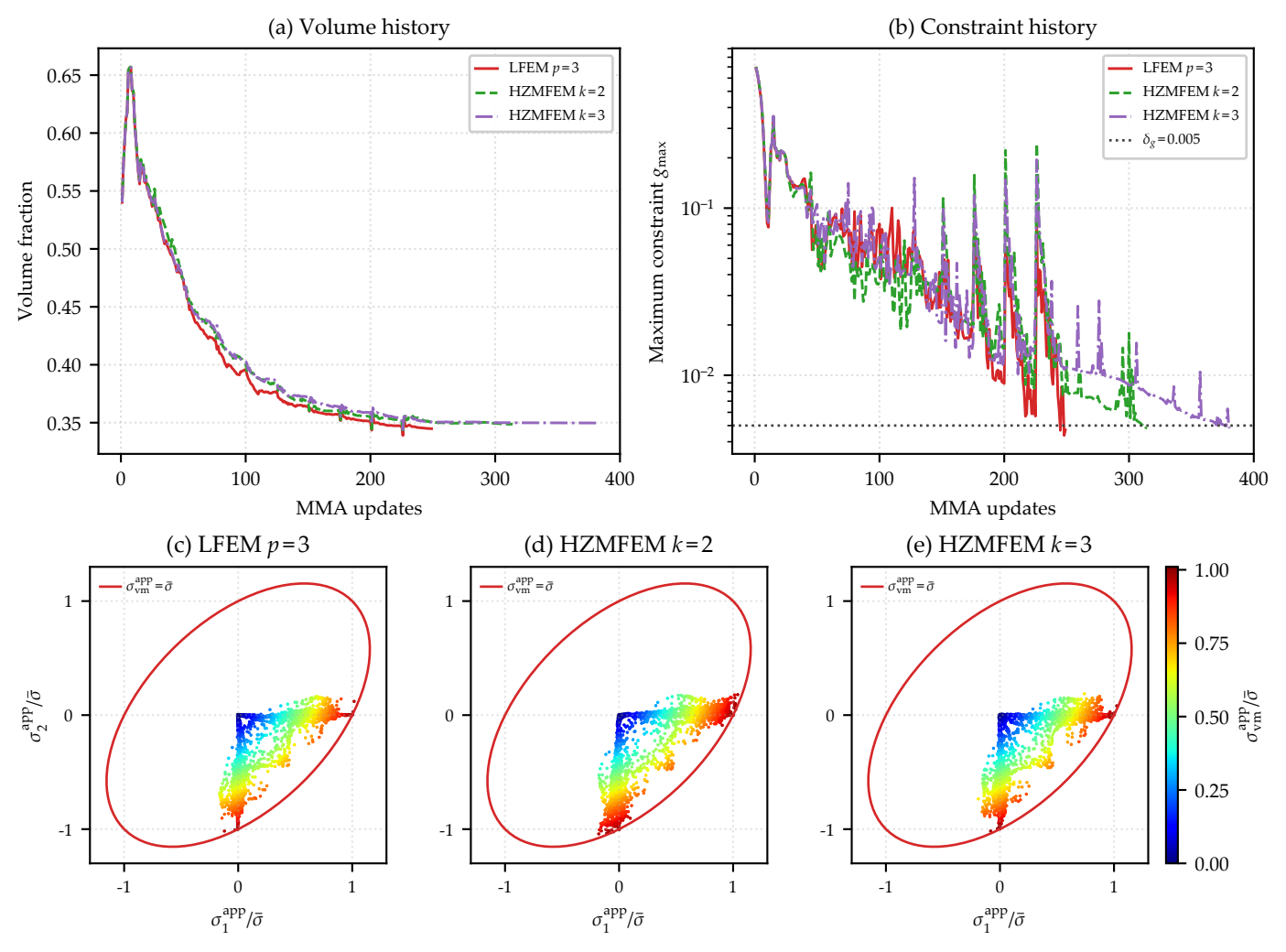

(a) Volume history
Volume fraction
MMA updates
LFEM $p=3$
HZMFEM $k=2$
HZMFEM $k=3$
(b) Constraint history
Maximum constraint $g_{\max}$
MMA updates
LFEM $p=3$
HZMFEM $k=2$
HZMFEM $k=3$
$\delta_g=0.005$
(c) LFEM $p=3$
(d) HZMFEM $k=2$
(e) HZMFEM $k=3$
$\sigma_{\mathrm{vm}}^{\mathrm{app}}=\bar{\sigma}$
$\sigma_1^{\mathrm{app}}/\bar{\sigma}$
$\sigma_2^{\mathrm{app}}/\bar{\sigma}$
$\sigma_{\mathrm{vm}}^{\mathrm{app}}/\bar{\sigma}$